

CSE-719(Distributed and Cloud Computing)- Distributed Systems

Dr. Atiqur Rahman

ড. আতিকুর রহমান

Ph.D.(CQUPT, China), MS.Engg.(CU), B.Sc.(CU)

Associate Professor

Department of Computer Science and Engineering

University of Chittagong

Lecture 1: Welcome, and Introduction

Our Main Goal Today

To Define the Term Distributed System

Can you name some examples of
Operating Systems?

Can you name some examples of Operating Systems?

...

Linux WinXP Vista 7/8 Unix FreeBSD Mac OSX

2K Aegis Scout Hydra Mach SPIN

OS/2 Express Flux Hope Spring

AntaresOS EOS LOS SQOS LittleOS TINOS

PalmOS WinCE TinyOS iOS

...

What is an Operating System?

What is an Operating System?

- User interface to hardware (device driver)
- Provides abstractions (processes, file system)
- Resource manager (scheduler)
- Means of communication (networking)
- ...

FOLDOC definition

(FOLDOC = Free On-Line Dictionary of Computing)

Operating System-The low-level software which handles the interface to peripheral hardware, schedules tasks, allocates storage, and presents a default interface to the user when no application program is running.

Can you name some examples of
Distributed Systems?

Can you name some examples of Distributed Systems?

- Client-Server (NFS)
- The Web
- The Internet
- A wireless network
- DNS
- Gnutella or BitTorrent (peer to peer overlays)
- A “cloud”, e.g., Amazon EC2/S3, Microsoft Azure
- A datacenter, e.g., NCSA, a Google datacenter, The Planet

What is a Distributed System?

FOLDOC definition

A collection of (probably heterogeneous) automata whose distribution is transparent to the user so that the system appears as one local machine. This is in contrast to a network, where the user is aware that there are several machines, and their location, storage replication, load balancing and functionality is not transparent. Distributed systems usually use some kind of client-server organization.

Textbook definitions

- A distributed system is a collection of independent computers that appear to the users of the system as a single computer.

[Andrew Tanenbaum]

- A distributed system is several computers doing something together. Thus, a distributed system has three primary characteristics: multiple computers, interconnections, and shared state.

[Michael Schroeder]

Unsatisfactory

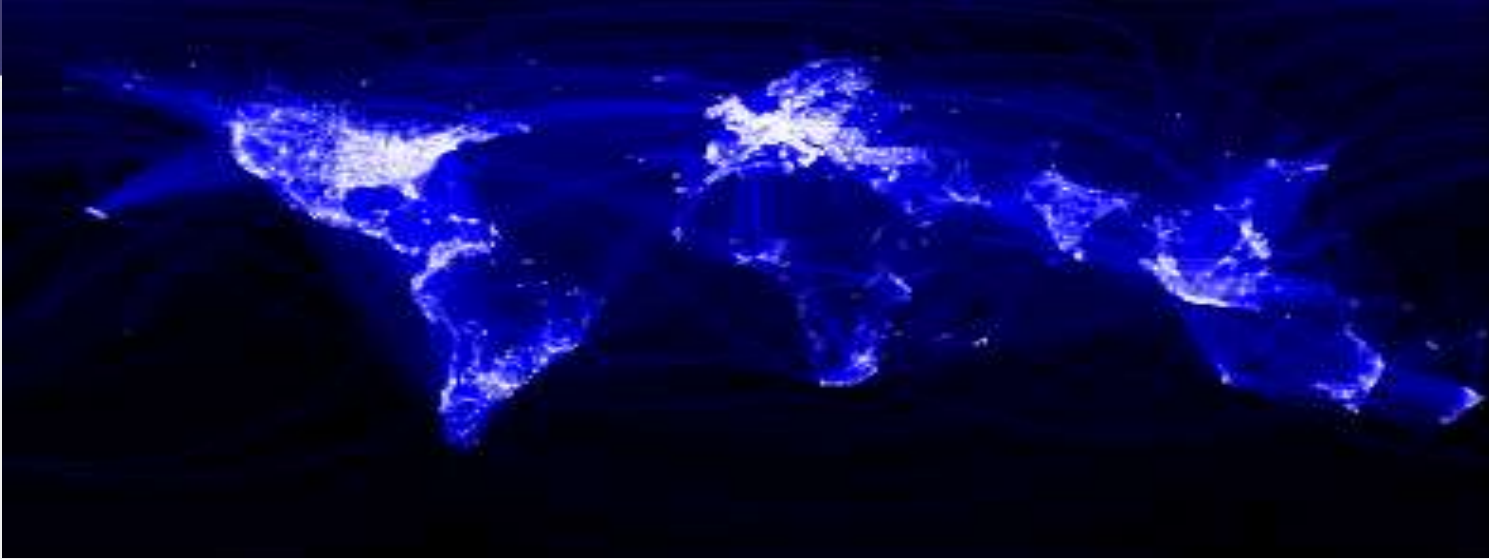
- Why are these definitions short?
- Why do these definitions look inadequate to us?
- Because we are interested in the insides of a distributed system
 - design and implementation
 - Maintenance
 - Algorithmics (“protocols”)

“I shall not today attempt further to define the kinds of material I understand to be embraced within that shorthand description; and perhaps I could never succeed in intelligibly doing so. But I know it when I see it, and the motion picture involved in this case is not that.”

[Potter Stewart, Associate Justice, US Supreme Court (talking about his interpretation of a technical term laid down in the law, case Jacobellis versus Ohio 1964)]

Which is a Distributed System – (A) or (B)?

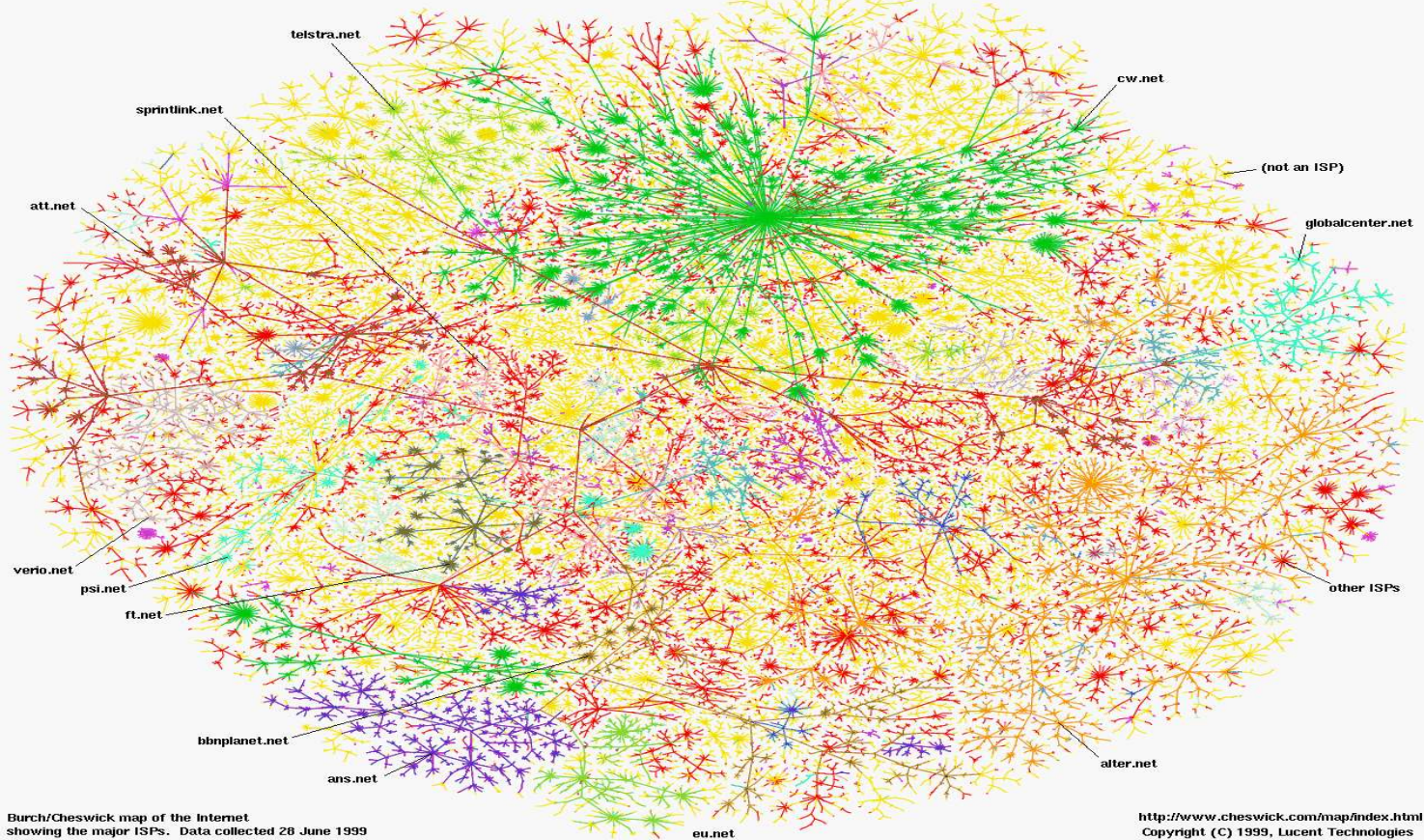
(A)



(A) Facebook Social Network Graph among humans

Source: https://www.facebook.com/note.php?note_id=469716398919

(B)



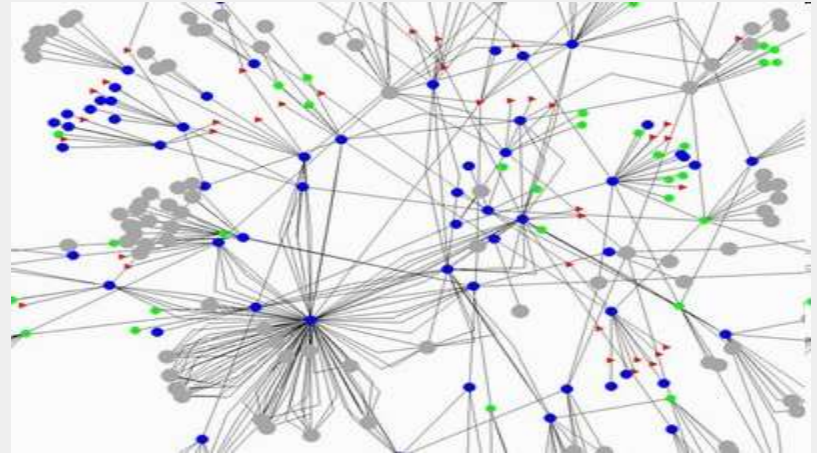
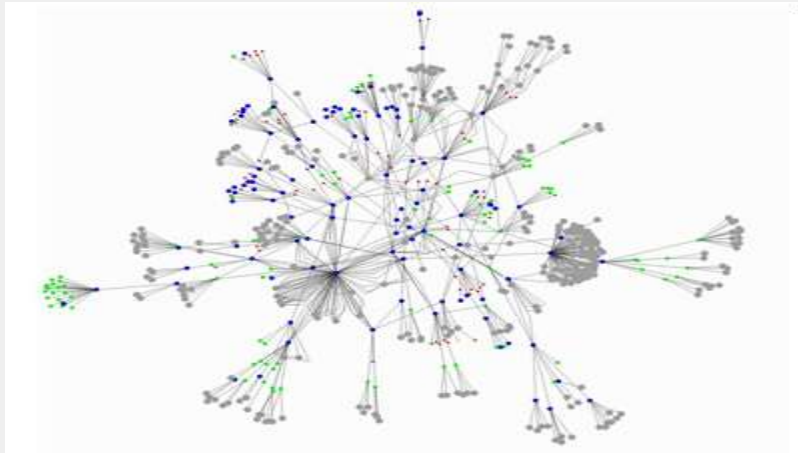
(B) The Internet (Internet Mapping Project, color coded by ISPs)

A working definition for us

*A distributed system is a collection of entities, each of which is **autonomous**, **programmable**, **asynchronous** and **failure-prone**, and which communicate through an **unreliable** communication medium.*

- Entity=a process on a device (PC, PDA)
- Communication Medium=Wired or wireless network
- Our interest in distributed systems involves
 - design and implementation, maintenance, algorithmic

Gnutella Peer to Peer System



What are the “entities”
(nodes)?

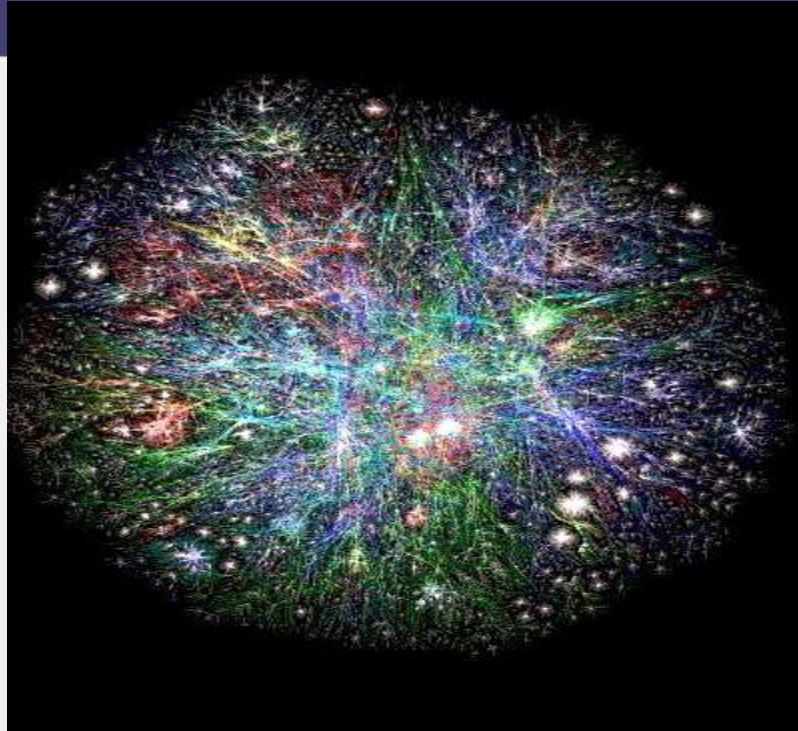
Source: GnuMap Project

What is the
communication medium
(links)?

Web Domains

What are the “entities”
(nodes)?

What is the
communication medium
(links)?



Source: <http://www.vlib.us/web/worldwideweb3d.html>

Datacenter



What are the “entities”
(nodes)?

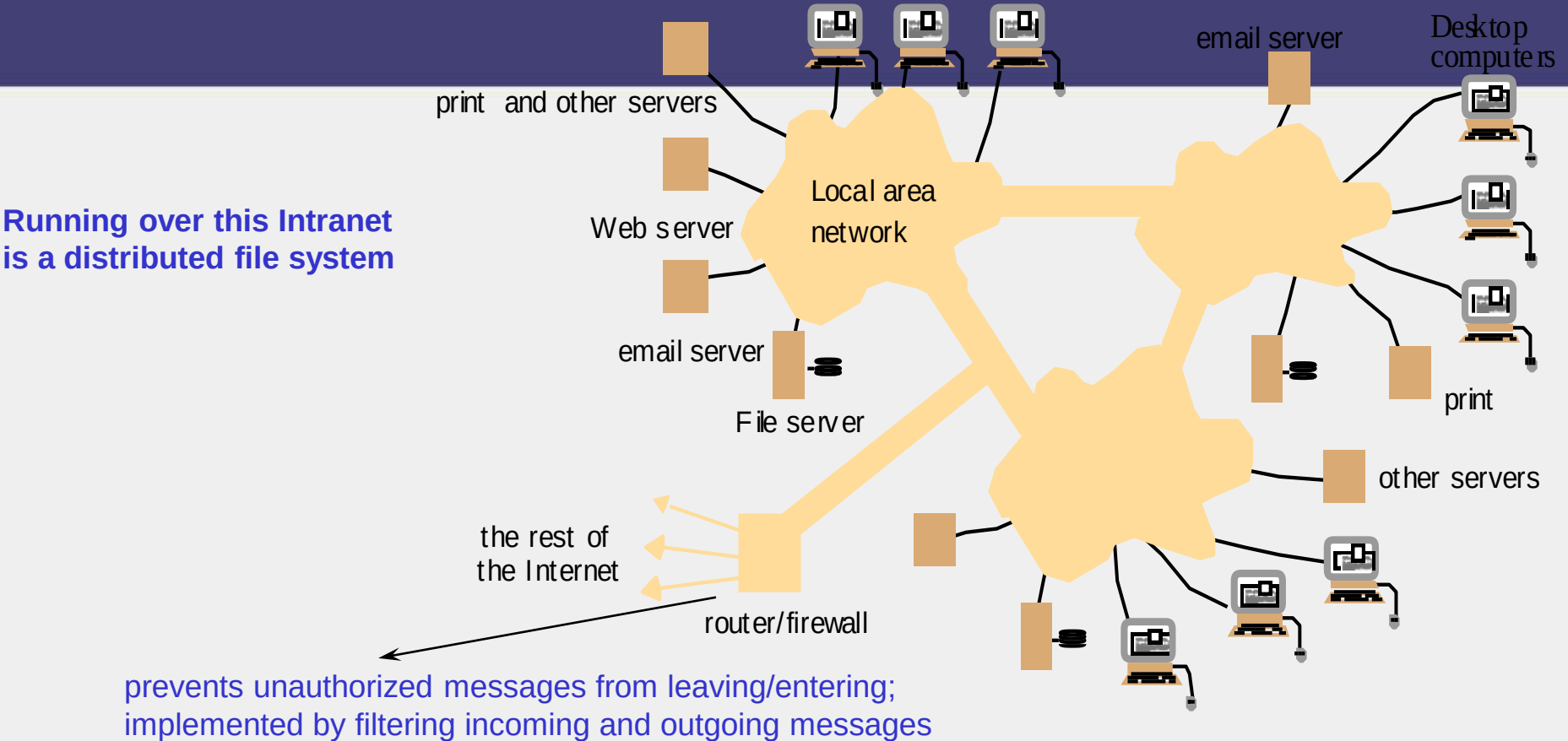
What is the
communication medium
(links)?



The Internet – Quick Refresher

- Underlies many distributed systems.
- A vast interconnected collection of computer networks of many types.
- Intranets – subnetworks operated by companies and organizations.
- Intranets contain subnets and LANs.
- WAN – wide area networks, consists of LANs
- ISPs – companies that provide modem links and other types of connections to users.
- Intranets (actually the ISPs' core routers) are linked by backbones – network links of large bandwidth, such as satellite connections, fiber optic cables, and other high-bandwidth circuits.
- UC2B? Google Fiber?

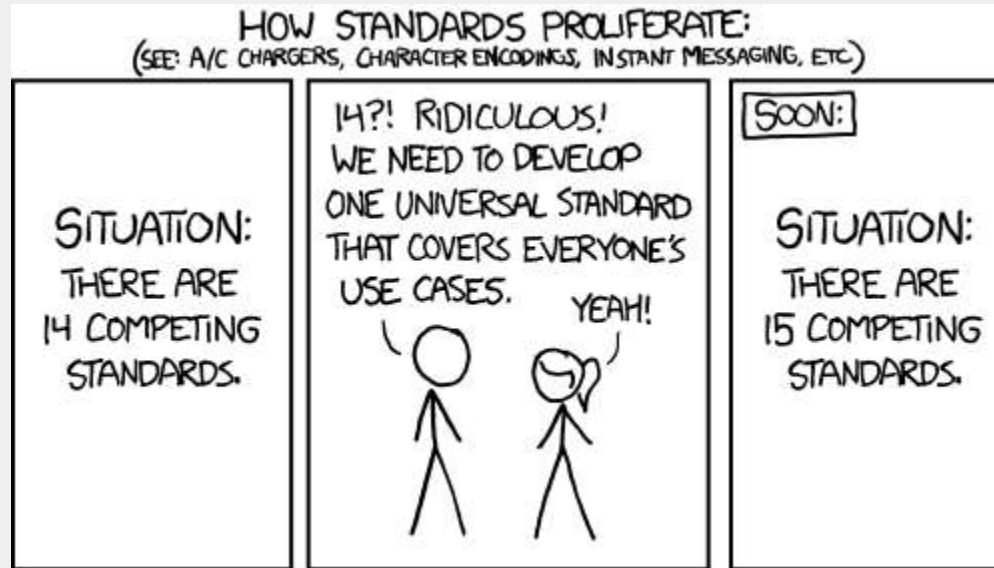
An Intranet & a distributed system



Networking Stacks

Application	Application layer protocol	Underlying transport protocol
Distributed System Protocols!		Networking Protocols
e-mail	smtp [RFC 821]	TCP
remote terminal access	telnet [RFC 854]	TCP
Web	http [RFC 2068]	TCP=Transmission Control Protocol UDP=User Datagram Protocol (Implemented via sockets)
file transfer	ftp [RFC 959]	
streaming multimedia	proprietary (e.g. RealNetworks)	
remote file server	NFS	TCP or UDP
Internet telephony	proprietary (e.g., Skype)	typically UDP

The History of Internet Standards

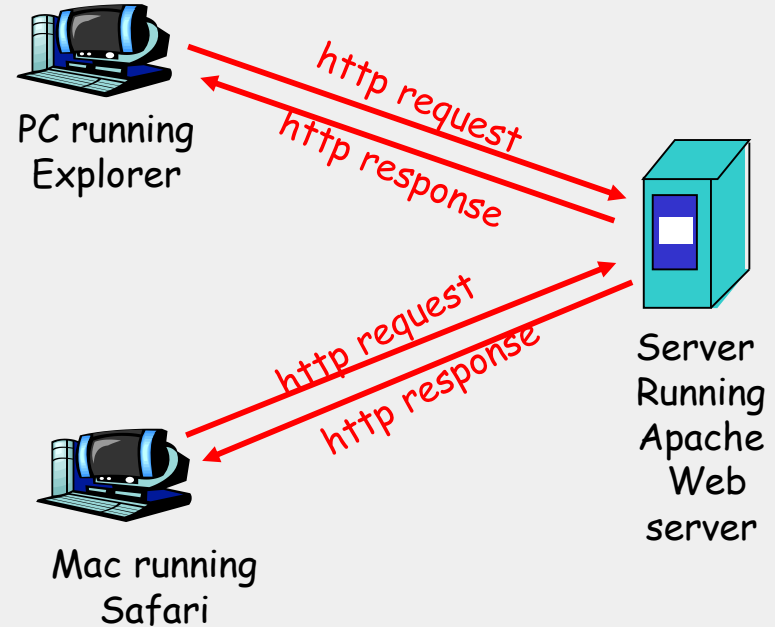


Source: <http://xkcd.com/927/>

The Heart of the World Wide Web: the HTTP Standard

HTTP: hypertext transfer protocol

- WWW's application layer protocol
- client/server model
 - *client*: browser that requests, receives, and “displays” WWW objects
 - *server*: WWW server, which is storing the website, sends objects in response to requests
- http1.0: RFC 1945
- http1.1: RFC 2068
 - Leverages same connection to download images, scripts, etc.



The HTTP Protocol: More

http: TCP transport service:

- client initiates a TCP connection (creates socket) to server, port 80
- server accepts the TCP connection from client
- http messages (application-layer protocol messages) exchanged between browser (http client) and WWW server (http server)
- TCP connection closed

http is “stateless”

- server maintains no information about past client requests

Why?

Protocols that maintain session “state” are complex!

- past history (state) must be maintained and updated.
- if server/client crashes, their views of “state” may be inconsistent, and hence must be reconciled.
- RESTful protocols are stateless.

HTTP Example

Suppose user enters URL `www.cs.uiuc.edu/`

(contains text,
references to 10
jpeg images)

1a. http client initiates a TCP connection to
http server (process) at
`www.cs.uiuc.edu`. Port 80 is default for
http server.

1b. http server at host `www.cs.uiuc.edu`
waiting for a TCP connection at port
80. “accepts” connection, notifying
client

2. http client sends a http *request message*
(containing URL) into TCP
connection socket

3. http server receives request messages,
forms a *response message* containing
requested object (`index.html`), sends
message into socket

time



HTTP Example (cont.)

5. http client receives a response message containing html file, displays html, Parses html file, finds 10 referenced jpeg objects

6. *Steps 1-5 are then repeated for each of 10 jpeg objects*

4. http server closes the TCP connection (if necessary).

time ↓ For fetching referenced objects, have 2 options:

- **non-persistent connection:** only one object fetched per TCP connection
 - some browsers create multiple TCP connections *simultaneously* - one per object
- **persistent connection:** multiple objects transferred within one TCP connection

Your Shell as a browser

1. Telnet to your favorite WWW server:

```
telnet www.google.com 80
```

Opens TCP connection to port 80
(default http server port) at www.google.com
Anything typed in sent
to port 80 at www.google.com

2. Type in a GET http request:

```
GET /index.html
```

Or

```
GET /index.html HTTP/1.0
```

By typing this in (may need to hit
return twice), you send
this minimal (but complete)
GET request to http server

3. Look at response message sent by http server!

What do you think the response is?

Does our Working Definition work for the http Web?

*A distributed system is a collection of entities, each of which is **autonomous**, **programmable**, **asynchronous** and **failure-prone**, and that communicate through an **unreliable** communication medium.*

- Entity=a process on a device (PC, PDA)
- Communication Medium=Wired or wireless network
- Our interest in distributed systems involves
 - design and implementation, maintenance, study, algorithmic

“Important” Distributed Systems Issues

- No global clock; no single global notion of the correct time (asynchrony)
- Unpredictable failures of components: lack of response may be due to either failure of a network component, network path being down, or a computer crash (failure-prone, unreliable)
- Highly variable bandwidth: from 16Kbps (slow modems or Google Balloon) to Gbps (Internet2) to Tbps (in between DCs of same big company)
- Possibly large and variable latency: few ms to several seconds
- Large numbers of hosts: 2 to several million

Many Interesting Design Problems

-
-
- Real distributed systems
 - Cloud Computing, Peer to peer systems, Hadoop, key-value stores/NoSQL, distributed file systems, sensor networks, measurements, graph processing, stream processing, ...
- Classical Problems
 - Failure detection, Asynchrony, Snapshots, Multicast, Consensus, Mutual Exclusion, Election, ...
- Concurrency
 - RPCs, Concurrency Control, Replication Control, ...
- Security
 - Byzantine Faults, ...
- Others...
-

Typical Distributed Systems Design Goals

- Common Goals:

- **Heterogeneity** – can the system handle a large variety of types of PCs and devices?
- **Robustness** – is the system resilient to host crashes and failures, and to the network dropping messages?
- **Availability** – are data+services always there for clients?
- **Transparency** – can the system hide its internal workings from the users?
- **Concurrency** – can the server handle multiple clients simultaneously?
- **Efficiency** – is the service fast enough? Does it utilize 100% of all resources?
- **Scalability** – can it handle 100 million **nodes** without degrading service?
(nodes=clients and/or servers) How about 6 B? More?
- Security – can the system withstand hacker attacks?
- **Openness** – is the system extensible?

“Important” Issues

- If you’re already complaining that the list of topics we’ve discussed so far has been perplexing...
 - You’re right!
 - It was meant to be (perplexing)
- The Goal for the Rest of the Course: see enough **examples** and learn enough **concepts** so these topics and issues will make sense
 - We will revisit many of these slides in the very last lecture of the course!

“Concepts”?

- Which of the following inventions do you think is the most important?
 1. Car
 2. Wheel
 3. Bicycle

“What lies beneath?” Concepts!

How will We Learn?

- Textbook, Recommended but not Limited to
 - Distributed and Cloud Computing- From Parallel Processing to the Internet of Things by Kai Hwang, Geoffery C. Fox, Jack J. Dongarra (Any edition)
 - Fundamentals of Cloud Computing by A. Kannammal
 - Cloud Computing by Michael Miller
- Lectures
- Homework's
 - Approx. one every two-three weeks
 - Solutions need to be typed, figures can be hand-drawn
 - May have extra problems for 4-credit students
- Programming assignments (3-4)
- Exams/quizzes
 - CTS' + Final

What assistance is available to you?

- Lectures
 - lecture slides will be placed online at Google Classroom or Students' Group
 - “Tentative” version before lecture
 - “Final” version after lecture
- Homework's – office hours and discussion forum to help you (without giving you the solution).
- Programming Assignments (MPs) – First two assignments will be in C
 - Office hours and Piazza discussion forum to help you (without giving you the solution).
- Course Prerequisite: CSE511 or CSE613 or equivalent OS/Networking course (latter need instructor permission)

You can find us almost everywhere

In person:

- Myself (*Dr. Atiq*): Associate Professor, CSE department, CU.
 - Office Hours Every Tuesday and Thursday right after class.
- TAs:

(Check Google Classroom or Group for office hours. There is > 0 office hours every day of the week.)

Virtually:

- Discussion Forum: Piazza (most preferable, monitored daily) – 24 hour turnaround time for questions!
- Email (turnaround time may be longer than Piazza) – use atiqcse09@cu.ac.bd
- Email individuals (instructor, TA) only if absolutely necessary (e.g., private matter)

Wrap-Up

- (Reading for today' s lecture: Relevant parts of Chapter 1)
- All students: Must join the Google Classroom and Group ASAP.
- Next lecture
 - Topic: “Introduction to Cloud Computing”